

KYZYL, Russian Federation

WMO: 360960

Lat: 51.717N

Lon: 94.500E

Elev: 2074

StdP: 13.63

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-40.4	-36.9	-44.7	0.4	-40.3	-41.3	0.5	-36.9	7.6	-6.9	6.0	-10.7	0.8	60	0.568

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.6	90.9	63.1	86.9	62.2	83.3	61.1	66.3	82.2	64.7	80.4	63.3	78.3	4.3	30

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
61.4	87.9	70.2	59.6	82.2	69.3	57.8	76.9	68.9	32.3	82.0	31.0	80.3	29.9	78.7	72.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 14.0	11.3	9.2	DB	-41.7	97.3	4.6	3.1	-45.1	99.5	-47.8	101.3	-50.4	103.0	-53.8	105.3	(4)
(5)			WB	-41.9	68.6	4.6	2.0	-45.1	70.1	-47.8	71.2	-50.4	72.3	-53.7	73.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	31.2	-19.7	-9.1	16.8	42.7	53.7	66.7	70.3	64.8	52.0	35.7	11.3	-12.9		
	DBStd	32.75	9.70	11.70	13.62	9.29	8.18	7.08	6.05	6.41	7.49	7.95	12.34	11.49		
	HDD50	8770	2162	1655	1030	250	55	0	0	0	63	445	1160	1950		
	HDD65	12717	2627	2075	1495	670	359	64	19	83	393	907	1610	2415		
	CDD50	1918	0	0	0	30	171	502	629	459	124	3	0	0		
	CDD65	390	0	0	0	0	10	116	183	77	4	0	0	0		
	CDH74	4349	0	0	0	12	259	1378	1795	823	82	0	0	0		
	CDH80	1587	0	0	0	0	62	516	703	289	17	0	0	0		
	WSAvg	3.2	1.6	2.0	3.1	4.6	5.1	4.5	3.9	3.7	3.5	3.0	2.1	1.6		
	PrecAvg	10.2	0.4	0.2	0.2	0.4	0.6	1.5	2.1	2.0	1.2	0.4	0.6	0.5		
	PrecMax	13.1	0.6	0.5	0.6	0.7	1.5	2.6	4.6	3.3	3.0	1.2	1.3	0.9		
(15) Precipitation	PrecMin	7.8	0.2	0.1	0.1	0.0	0.2	0.4	0.8	0.7	0.4	0.1	0.1	0.2		
	PrecStd	1.7	0.1	0.1	0.1	0.2	0.3	0.6	1.1	0.7	0.6	0.3	0.3	0.2		
	DB	11.7	22.5	51.5	75.3	86.4	94.5	96.7	93.9	82.2	64.4	41.7	20.5			
	MCWB	10.6	19.3	39.7	51.1	58.1	63.4	65.7	63.7	59.6	48.3	34.9	18.8			
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	2% DB	3.6	16.9	46.2	69.0	80.9	89.7	91.8	87.2	75.2	57.6	36.6	12.8			
	2% MCWB	2.5	14.6	37.0	48.7	55.2	61.7	64.8	63.2	56.7	44.9	31.8	11.7			
	5% DB	-1.1	12.4	41.5	64.3	76.2	85.9	87.3	82.4	70.4	52.7	32.2	8.6			
	5% MCWB	-2.0	10.5	33.9	46.0	53.0	61.0	63.4	62.2	54.7	42.5	28.7	7.6			
	10% DB	-5.1	8.4	36.0	59.2	70.9	82.1	83.5	78.1	66.1	48.4	28.3	3.4			
	10% MCWB	-5.7	7.0	30.8	44.1	50.8	59.6	62.3	60.5	52.7	39.8	25.5	2.6			
	0.4% WB	10.7	19.7	40.4	52.8	59.3	66.5	69.5	67.0	61.0	49.0	36.2	18.4			
	0.4% MCDB	12.0	22.4	49.5	71.5	82.3	84.5	87.1	81.5	77.7	62.3	40.0	20.4			
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2% WB	2.5	14.5	37.5	49.5	56.5	64.1	67.2	65.0	58.2	45.7	32.1	11.7			
	2% MCDB	3.5	16.7	45.8	66.2	77.5	82.3	82.7	80.5	72.6	56.4	36.2	12.7			
	5% WB	-1.9	10.7	34.3	47.0	54.2	62.7	65.6	63.5	55.9	43.0	29.0	7.6			
	5% MCDB	-1.0	12.4	40.5	62.0	72.5	79.7	81.1	78.3	68.2	51.8	32.1	8.5			
	10% WB	-5.7	7.0	31.0	44.8	51.8	61.3	64.2	61.9	53.6	40.5	25.6	2.7			
	10% MCDB	-5.1	8.4	36.0	58.3	68.5	78.3	79.1	75.5	64.2	47.4	28.2	3.3			
	MDBR	14.5	20.7	22.2	22.3	23.3	23.0	20.6	21.0	22.6	20.3	14.8	12.5			
	5% DB	15.3	21.1	22.2	30.7	33.1	29.5	27.4	28.9	29.3	27.6	17.4	12.5			
(36) Mean Daily Temperature Range	MCWBR	14.5	19.0	13.8	15.0	13.7	10.2	8.9	10.7	13.3	15.9	12.8	11.5			
	MCDBR	15.2	21.2	21.4	27.4	29.3	24.4	21.6	24.0	25.9	26.3	16.4	12.4			
	MCWBR	14.4	19.1	13.6	14.0	12.9	9.0	8.1	9.5	11.9	15.5	12.1	11.5			
	MCDBR	14.4	19.1	13.6	14.0	12.9	9.0	8.1	9.5	11.9	15.5	12.1	11.5			
(40) Clear-Sky Solar Irradiance	taub	0.265	0.295	0.341	0.381	0.395	0.389	0.389	0.375	0.336	0.320	0.297	0.271			
	taud	2.128	2.083	2.067	2.202	2.223	2.345	2.377	2.384	2.435	2.375	2.210	2.102			
	Ebn at Noon	231	256	266	270	272	274	272	271	270	249	216	203			
	Edh at Noon	23	33	41	41	42	37	36	34	29	25	22	20			
(44) All-Sky Solar Radiation	RadAvg	426	787	1311	1591	1800	1997	1829	1572	1202	744	404	309			
	RadStd	17	32	47	76	113	146	108	95	113	50	33	16			

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (7 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page